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FREE VIBRATION ANALYSIS OF ROTATING STIFFENED COMPOSITE CYLINDRICAL SHELLS BY USING THE LAYERWISE-DIFFERENTIAL QUADRATURE (LW-DQ) METHOD

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This paper focuses on an analysis of free vibration of thick rotating stiffened composite cylindrical shells with different boundary conditions. The analysis is performed on the basis of a three-dimensional theory by using the layerwise-differential quadrature method (LW-DQM). The equations of motion are derived employing Hamilton's principle. In order to accurately allow for the thickness effects, a layerwise theory is used to discretize the equations of motion and related boundary conditions through the thickness of the shells. Then, the equations of motion and the boundary conditions are transformed into a set of algebraic equations by using the DQM in the longitudinal direction. This study demonstrates the applicability, accuracy, stability, and fast rate of convergence of the present method in free vibration analyses of rotating stiffened cylindrical shells. The presented results are compared with those of other shell theories obtained by conventional methods and with a special case where the number of stiffeners approaches zero, i.e., an nonstiffened cylindrical shell, and excellent agreements are achieved. Finally, some new results are presented, which can be used as benchmark solutions for future investigations.

1. Introduction

Rotating orthogonally stiffened cylindrical shells are increasingly being used in many engineering applications, such as the drive shafts of gas turbines, high-speed centrifugal separators, motors, and rotor systems, because of their high

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strength to weight ratio. Hence, it is of a great importance to understand the vibration behavior of such shells for the design of aforementioned structures. Most studies were restricted to a vibration analysis of nonrotating and rotating cylindrical shells based on the classical theory of laminated shells (CTLS). They include the works by Lam and Loy on rotating composite and sandwich-type cylindrical shells [1, 2], as well as a comparison study on different thin shell theories, and also a discussion on the influence of boundary conditions of rotating cylindrical shells [3, 4]. A simplified theory for analyzing the first critical speed of a composite cylindrical shell was used by Kim and Bert [5]. The vibration and critical speed of thin, isotropic cylindrical shells under constant axial loads was studied by Ng and Lam [6]. Their theory cannot be used for thick and even thin shells, when the number of circumferential waves increases as a result of neglecting shear deformations and rotary inertia effects in the CTLS. The refinement of thin-shell theories has resulted in a number of the so-called first- and higher-order shear deformation theories to include the effects of transverse shear deformations [7-14]. These theories are much more accurate than the thin-shell theories in an analysis of moderately thick shells, but they are still inadequate for an analysis of thick and stiffened shells. In order to analyze thick and/or stiffened cylindrical shells accurately, a three-dimensional theory should be used to take into account all transverse stress and strain components. A dynamic analysis and exact solution for the vibration of rotating cylindrical shells based on a 3-D theory is complex, and numerical methods, such as the finite-element, Chebyshev-Ritz, and series solution methods, are most often used in these investigations [15-21].

In recent years, a new numerical method, the so-called hybrid layerwise-differential quadrature method (LW-DQM), has been employed to discretize the governing equations of structures [22-25]. In contrast to the equivalent single-layer theories (ESLT), the layerwise theories develop separate expansions of displacement fields for each subdivision. The displacement components are continuous through the laminate thickness, but their derivatives with respect to the thickness coordinate may be discontinuous at various points across through the thickness, thereby allowing for the possibility of continuous transverse stresses at the interfaces separating contradictory materials [26].

The differential quadrature method (DQM) is an efficient numerical technique, which was originated by Bellman [27] to solve linear and nonlinear differential equations. The basic idea of the method is that the partial derivative of a function with respect to a variable at a given point can be approximated as a weighted linear sum of values of the function at all discrete points in the domain of that variable. The mathematical fundamentals and recent developments of the DQ method, as well as its major applications in engineering, are discussed by Shu [28]. It is worth noting that the interest of researchers [29-36] in this procedure is increasing due to its great simplicity and versatility.

The vibration analysis of stiffened cylindrical shells has been and continues to be of considerable interest to structural analysts because of the widespread use of this or similar structures. The investigative efforts may be divided into two broad classes:

a) those that consider the stiffeners to be closely spaced and average or “smear” the stiffening effects over the entire surface of the shell, thus effectively replacing the stiffened shell by an orthotropic one;

b) those that do not consider the stiffeners to be closely spaced and do not take advantage of the simplification of averaging the stiffener effects; moreover the stiffeners may be spaced arbitrarily and need not be identical. In [37-38], the averaging technique was applied to the analysis of stiffened shells, but the discrete approach was used in [39-42].

The literature clearly shows that there have been no studies on the vibration of rotating stiffened cylindrical shells with arbitrary boundary conditions by using the three-dimensional theory of elasticity, in spite of the fact that the critical rotational speed and the vibration of stiffened cylindrical shells is important to their applications. Hence, in this study, based on a 3-D theory, a hybrid LW-DQ method for a free vibration analysis of rotating stiffened composite cylindrical shell with arbitrary boundary conditions is developed. It should be noted that the problem considered in this paper is of dual purpose: first, to accomplish the frequency analysis for a thick-walled cylindrical shell made of a layered composite that is limited in length and is rotated around its symmetrical and horizontal axis, and, second, to present a semianalytical method for a three-dimensional solution of the boundary dynamic problem in cylindrical coordinates. The accuracy, convergence, and versatility of the algorithm are demonstrated by different examples both for nonstiffened and stiffened shells. Also, the effects of the depth to width ratio and the number of stiffeners are shown.

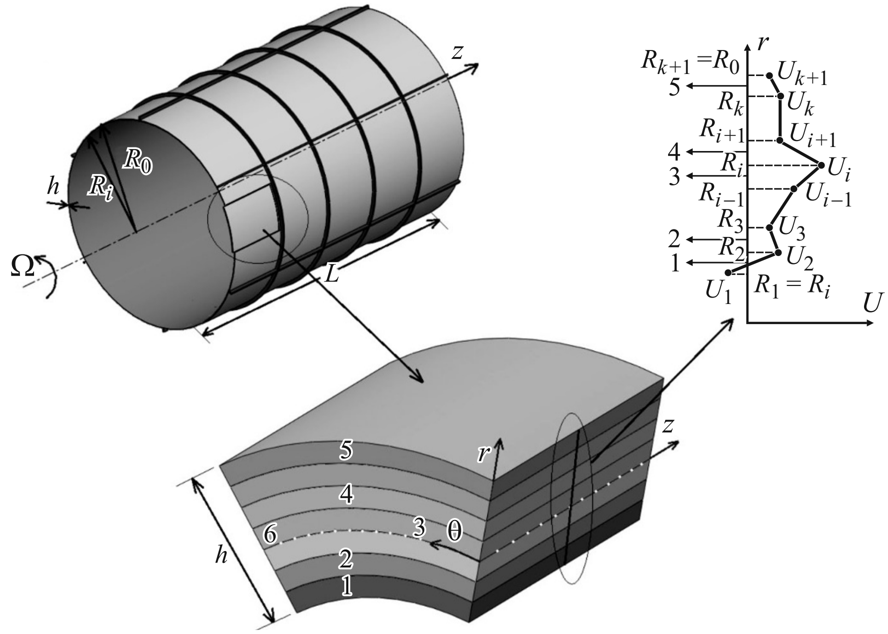

Fig. 1. Geometry of the structure of a rotating stiffened cylindrical shell: 1, 2, 3, 4, and 5 — the first, second, $(i-1)$th, i-th, and k-th layers; 6 — reference surface..

2. Problem Formulation

2.1. Geometrical configuration

The rotating stiffened cylindrical shell is considered to be thick, laminated, and composed of an arbitrary number of layers, as shown in Fig. 1, where L is the length, h is the thickness, R_i and R_o are the inner and outer radii, and Ω is the constant angular speed of the cylindrical shell about its symmetrical horizontal axis. The reference surface of the cylindrical shell is taken to be its middle surface, where an orthogonal coordinate system (r, θ, z) is fixed, and the displacement of the shell in the r, θ, and z directions are denoted by w, v, and u, respectively. The depth and width of the stiffeners are symbolized by $d_{s(\text{or } r)}$ and $b_{s(\text{or } r)}$. The subscripts s and r indicate the stringer and ring stiffeners, respectively. The displacement from the outer surface of shell to any point located on the stringer and ring are denoted by z_s and z_r, respectively.

2.2. Displacement field in the layerwise theory

The displacement field of closed cylindrical shells in the circumferential direction can be expanded as

$$u(r,\theta,z,t)=\sum_{n=1}^{\infty}U_n(r,z)\cos(n\theta+\omega t),$$

$$v(r,\theta,z,t)=\sum_{n=1}^{\infty}V_n(r,z)\sin(n\theta+\omega t),$$

$$w(r,\theta,z,t)=\sum_{n=1}^{\infty}W_n(r,z)\cos(n\theta+\omega t),$$

where n is the circumferential wave number.

In order to build a high-degree transverse discretization into the model, the layerwise laminate theory of Reddy [26] is extended to be used here. The layerwise concept of Reddy [11] is very general in that the number of subdivisions through the thickness can be greater than, equal to, or less than the number of material layers, and the pattern of layerwise interpolation through the thickness is very easy to include into the finite-element method. In this theory, the variation of the displacements U_n, V_n, and W_n through the thickness can be represented, to a certain desired level of accuracy, by simply increasing the order of transverse interpolation polynomials:

$$U_n(r,z) = \sum_{i=1}^{N_{nr}} U_{in}(z)\psi_i(r),$$

$$V_n(r,z) = \sum_{i=1}^{N_{nr}} V_{in}(z)\psi_i(r),$$

$$W_n(r,z) = \sum_{i=1}^{N_{nr}} W_{in}(z)\psi_i(r),$$

where $U_{in}(z)$, $V_{in}(z)$, and $W_{in}(z)$ are the unknown displacement components at an ith node in the z, θ, and r directions, respectively; N_{nr} is the number of nodes in the r direction of the shell, which depends on the number of considered mathematical layers N_l through the thickness and the number of nodes per layer N_{pl} in the thickness direction as $N_{nr} = N_l\left(N_{pl} - 1\right) + 1$. Also, $\psi_i(r)$ is a piecewise continuous Lagrange interpolation function in terms of the thickness coordinate r in an ith mathematical layer and is assumed to take the following form:

$$\psi_i(r) = \begin{cases} 0, & r \le R_{i-1}, \\ \dfrac{r - R_{i-1}}{h_{i-1}}, & R_{i-1} \le r \le R_i, \\ \dfrac{R_{i+1} - r}{h_i}, & R_i \le r \le R_{i+1}, \\ 0, & r \ge R_{i+1,} \end{cases}$$

where h_i is the thickness of the ith layer, and R_i denotes the inner radius of this layer. Due to the independent interpolation through the thickness, this layerwise theory reduces the 3-D problem to a 2-D one.

2.3. Energy equations of the shell

The strain energy of the laminated composite cylindrical shell is expressed as

$$U_\varepsilon^{sh} = \frac{1}{2}\int_0^L \int_0^{2\pi} \int_{R_i}^{R_o} \varepsilon^T \left[\bar{C}\right] \varepsilon \, r dr d\theta \, dz, \tag{1}$$

where the strain vector ε, based on the three-dimensional theory of elasticity, can be written as

$$\varepsilon^T = \left\{\varepsilon_z, \varepsilon_\theta, \varepsilon_r, \gamma_{\theta r}, \gamma_{zr}, \gamma_{z\theta}\right\}.$$

The linearized strain–displacement relation is

$$\varepsilon = D\chi,$$

where the differential operator D and the displacement vector χ are defined as

$$D=\begin{bmatrix} \frac{\partial}{\partial z} & 0 & 0 \\ 0 & \frac{1}{r}\cdot\frac{\partial}{\partial\theta} & \frac{1}{r} \\ 0 & 0 & \frac{\partial}{\partial r} \\ 0 & \frac{\partial}{\partial r}-\frac{1}{r} & \frac{1}{r}\cdot\frac{\partial}{\partial\theta} \\ \frac{\partial}{\partial r} & 0 & \frac{\partial}{\partial z} \\ \frac{1}{r}\cdot\frac{\partial}{\partial\theta} & \frac{\partial}{\partial z} & 0 \end{bmatrix},\quad \chi^T=\{u, v, w\}.$$

$\left[\bar{C}\right]=\left[T\right]^{-1}\left[C\right]\left[T\right]$ is the transformed stiffness matrix for a laminated shell, where $\left[C\right]$ is the principal material stiffness matrix, and $\left[T\right]$ is the transformation matrix [11].

It should be noted that the work is done on the shell by the centrifugal force generated by rotation. This work can be written as

$$U_h^{sh}=\frac{1}{2}\int_0^L\int_0^{2\pi}\int_{R_i}^{R_o}N_\theta^{sh}\varepsilon_{\theta,Nl}^2 rdrd\theta dz, \tag{2}$$

where N_θ^{sh} and $\varepsilon_{\theta,Nl}$ are the initial hoop stress and the nonlinear circumferential strain due to the centrifugal force and are defined as [38]

$$N_\theta^{sh}=\rho\left(r\right)r^2\Omega^2,$$

$$\varepsilon_{\theta,Nl}^2=\left(\frac{1}{r}\cdot\frac{\partial u}{\partial\theta}\right)^2+\left[\frac{1}{r}\left(\frac{\partial v}{\partial\theta}+w\right)\right]^2+\left[\frac{1}{r}\left(\frac{\partial w}{\partial\theta}-v\right)\right]^2.$$

The kinetic energy of the rotating cylindrical shell is expressed as

$$T^{sh}=\frac{1}{2}\int_0^L\int_0^{2\pi}\int_{R_i}^{R_o}\rho\left(r\right)\mathbf{V}\cdot\mathbf{V}\,rdrd\theta dz. \tag{3}$$

Here $\mathbf{V}$ is the velocity vector at any point of the shell, which is given by

$$\mathbf{V}=\dot{\mathbf{r}}+\Omega\mathbf{k}\times\mathbf{r}, \tag{4}$$

where the point means differentiation with respect to time.

In Eq. (3), the displacement vector can be written as

$$\mathbf{r}=w\mathbf{i}+v\mathbf{j}+u\mathbf{k}, \tag{5}$$

where $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ denote the unit vectors in the r, θ, and z directions, respectively, in the nonrotating frame.

Substituting Eqs. (5) and (4) into Eq. (3), the kinetic energy of the shell can be presented in the form

$$T^{sh}=\frac{1}{2}\int_0^L\int_0^{2\pi}\int_{R_i}^{R_o}\rho(r)\begin{bmatrix}\left(\dot{u}\right)^2+\left(\dot{v}\right)^2+\left(\dot{w}\right)^2\\ +\Omega^2\left(v^2+w^2\right)+2\Omega\left(\dot{w}v-\dot{v}w\right)\end{bmatrix}rdrd\theta dz. \tag{6}$$

2.4. Energy equations of stiffeners

The stiffeners (rings and stringers) are assumed to be an integral part of the shell and are directly included in the analysis. In order to maintain the compatibility of displacements between the stiffeners and the shell, a special transformation is used, which includes the coupling effects caused by an eccentric placement of stiffeners. It should also be noted that the displacements vary through the stiffener depth. Therefore, the displacement of a point in the stiffener at a distance $z_{s(\mathrm{or}\ r)}$ from the outer surface of the shell can be expressed in the form

$$u_{s(or\,r)} = u_{N_{nr}} - z_{s(or\,r)} \frac{\partial w_{N_{nr}}}{\partial z},$$

$$v_{s(or\,r)} = v_{N_{nr}} - \frac{z_{s(or\,r)}}{R_o} \cdot \frac{\partial w_{N_{nr}}}{\partial \theta},$$

$$w_{s(or\,r)} = w_{N_{nr}}.$$

The strain of stringers in the longitudinal direction and the strain of rings in the circumferential direction are defined as

$$\varepsilon_{sz} = \frac{\partial u_{N_{nr}}}{\partial z} - z_s \frac{\partial^2 w_{N_{nr}}}{\partial z^2},$$

$$\varepsilon_{r\theta} = \frac{1}{R_o}\left(\frac{\partial v_{N_{nr}}}{\partial \theta} - \frac{z_r}{R_o} \frac{\partial^2 w_{N_{nr}}}{\partial \theta^2} + w_{N_{nr}} \right).$$

Using the theory of discrete stiffeners, the strain energy for the stringer can be written as

$$U_\varepsilon^s = \sum_{k=1}^{N_s} \frac{1}{2} E_{sk} \int_0^L \int_{A_{sk}} \varepsilon_{sz}^2 dA_{sk}\, dz + \sum_{k=1}^{N_s} \frac{1}{2} G_{sk} J_{sk} \int_0^L \frac{1}{R_o^2} \left(\frac{\partial^2 w}{\partial \theta\, \partial z} \right)^2 dz, \tag{7}$$

where $G_{sk}J_{sk}$ and A_{sk} are the torsional stiffness and the cross-sectional area of a kth stringer, respectively, and N_s is the number of stringers.

The strain energy of the ring can be written as

$$U_\varepsilon^r = \sum_{k=1}^{N_r} \frac{1}{2} E_{rk} \int_0^{2\pi} \int_{A_{rk}} \varepsilon_{r\theta}^2 dA_{rk}\, d\theta + \sum_{k=1}^{N_r} \frac{1}{2} G_{rk} J_{rk} \int_0^{2\pi} \frac{1}{R_o^2} \left(\frac{\partial^2 w}{\partial \theta\, \partial z} \right)^2 \left(R_o + \frac{d_r}{2} \right) d\theta, \tag{8}$$

where $G_{rk}J_{rk}$ and A_{rk} are the torsional stiffness and the cross-sectional area of a kth ring, respectively, and N_r is the number of rings.

The kinetic energy for the stringers and rings may be written as

$$T^s = \frac{1}{2} \rho_s \sum_{k=1}^{N_s} \int_0^L \int_{A_{sk}} \left[\dot{u}_s^2 + \dot{v}_s^2 + \dot{w}_s^2 + 2\Omega\left(v_s \dot{w}_s - w_s \dot{v}_s \right) + \Omega^2 \left(v_s^2 + w_s^2 \right) \right] dA_{sk}\, dz, \tag{9}$$

$$T^r = \frac{1}{2} \rho_r \sum_{k=1}^{N_r} \int_0^{2\pi} \int_{A_{rk}} \left[\begin{array}{l} \dot{u}_r^2 + \dot{v}_r^2 + \dot{w}_r^2 + 2\Omega\left(v_r \dot{w}_r - w_r \dot{v}_r \right) \\ + \Omega^2 \left(v_r^2 + w_r^2 \right) \end{array} \right] \left(R_o + \frac{d_s}{2} \right) dA_{rk}\, d\theta, \tag{10}$$

where $\rho_{s(or\,r)k}$ is the density of a kth stringer (or ring).

In the case of stringer, the hoop stress is not presented by the centrifugal force, but the work done on the ring is similar to that done on the shell. Therefore, the expressions of work for the stringer and ring are

$$U_h^s = 0, \tag{11}$$

$$U_h^r = \frac{1}{2} N_\theta^r \sum_{k=1}^{N_r} \int_0^{2\pi} \int_{A_{rk}} \left\{ \begin{array}{l} \left(\frac{1}{R_o}\frac{\partial u_s}{\partial \theta}\right)^2 + \left[\frac{1}{R_N}\left(\frac{\partial v_s}{\partial \theta} + w_s\right)\right]^2 \\ + \left[\frac{1}{R_0}\left(\frac{\partial w_s}{\partial \theta} - v_s\right)\right]^2 \end{array} \right\} \left(R_o + \frac{d_s}{2}\right) dA_{rk} d\theta. \tag{12}$$

2.5. Governing equations of motion

The governing differential equations of motion can be derived by using Hamilton's principle:

$$\int_{t_0}^{t_1} \delta(\Pi) dt = \int_{t_0}^{t_1} \left(\delta T^{sh} + \delta T^s + \delta T^r - \delta U_h^{sh} - \delta U_\varepsilon^{sh} - \delta U_\varepsilon^s - \delta U_\varepsilon^r - \delta U_h^r - \delta U_h^s\right) dt = 0. \tag{13}$$

Inserting Eqs. (1), (2), (6), and (7-12) into Eq. (13), we obtain the equations of motion and the related boundary conditions for a free vibration analysis of arbitrary rotating laminated cylindrical shell:

$$\begin{aligned}
\delta U_{in} : \pi \Bigg\{ \Bigg[& F_{55}^{ij} + n^2 C_{66}^{ij} - \left(\omega^2 + n^2\Omega^2\right) G^{ij} + \delta_{jN_{nr}} N_r \rho_r \Omega^2 R_r A_r n^2 \\
& - \frac{\delta_{jN_{nr}} \rho_s N_s A_s}{\pi} \omega^2 - \delta_{jN_{nr}} \rho_r N_r A_r \omega^2 \Bigg] U_{jn} \\
& - \Bigg[n\left(A_{12}^{ij} + A_{66}^{ij}\right) \frac{dV_{jn}}{dz} + \Bigg(A_{12}^{ij} + E_{13}^{ji} - E_{55}^{ij} + \frac{\delta_{jN_{nr}} N_r \rho_r \Omega^2 R_r A_r d_r n^2}{2} \\
& + \frac{\delta_{jN_{nr}} \rho_s N_s A_s d_s \omega^2}{2\pi} + \frac{\delta_{jN_{nr}} \rho_r N_r A_r d_r \omega^2}{2} \Bigg) \frac{dW_{jn}}{dz} \Bigg] \\
& + \left[-B_{11}^{ij} - \frac{\delta_{jN_{nr}} N_s E_s A_s}{\pi} \right] \frac{d^2 U_{jn}}{dz^2} - \left[- \frac{\delta_{jN_{nr}} N_s E_s A_s d_s}{2\pi} \right] \frac{d^3 W_{jn}}{dz^3} \Bigg\} = 0,
\end{aligned}$$

$$\begin{aligned}
\delta V_{in} : \pi \Bigg\{ \Bigg[& n^2 C_{22}^{ij} + F_{44}^{ij} - D_{44}^{ij} - D_{44}^{ji} + C_{44}^{ij} - B_{66}^{ij} \frac{\partial^2}{\partial z^2} - \left(\omega^2 + \Omega^2 n^2\right) G^{ij} \\
& + \delta_{jN_{nr}} N_r \frac{E_r A_r n^2}{R_r} + \delta_{jN_{nr}} N_r \rho_r \Omega^2 R_r A_r \left(n^2 + 1\right) \\
& - \frac{\delta_{jN_{nr}} \rho_s N_s A_s}{\pi} \left(\omega^2 + \Omega^2\right) - \delta_{jN_{nr}} \rho_r N_r A_r \left(\omega^2 + \Omega^2\right) \Bigg] V_{jn} \\
& + \Bigg[n\left(C_{22}^{ij} + D_{23}^{ji} + C_{44}^{ij} - D_{44}^{ij}\right) - 2\Omega\left(\omega + \Omega n\right) G^{ij} + \delta_{jN_{nr}} N_r \left(\frac{E_r A_r d_r n^3}{2R_r^2} + \frac{E_r A_r n}{R_r} \right)
\end{aligned}$$

$$+\delta_{jN_{nr}}\rho_r N_r A_r\left(2nR_r\Omega^2+\frac{d_r\left(n^3+n\right)\Omega^2}{2}-\frac{d_r n\omega^2}{2R_r}+2\Omega\omega-\frac{\Omega^2 d_r n}{2R_r}\right)$$

$$\left.\left.-\frac{\delta_{jN_{nr}}\rho_s N_s A_s}{\pi}\left(\frac{d_s n\omega^2}{2R_s}-2\Omega\omega+\frac{\Omega^2 d_s n}{2R_s}\right)\right]W_{jn}+n\left(A_{12}^{ij}+A_{66}^{ij}\right)\frac{d}{dz}U_{jn}\right\}=0,$$

$$\delta W_{in}:\pi\left\{\left[n\left(C_{22}^{ij}+D_{23}^{ij}+C_{44}^{ij}-D_{44}^{ji}\right)-2\Omega\left(\omega+\Omega n\right)G^{ij}+\frac{\delta_{jN_{nr}}N_r E_r A_r n}{R_r}\left(\frac{d_r n^2}{2R_r}+1\right)\right.\right.$$

$$+\delta_{jN_{nr}}\rho_r N_r A_r\left(2nR_r\Omega^2+\frac{d_r\left(n^3+n\right)\Omega^2}{2}-\frac{d_r n\omega^2}{2R_r}+2\Omega\omega-\frac{\Omega^2 d_r n}{2R_r}\right)$$

$$\left.-\frac{\delta_{jN_{nr}}\rho_s N_s A_s}{\pi}\left(\frac{d_s n\omega^2}{2R_s}-2\Omega\omega+\frac{\Omega^2 d_s n}{2R_s}\right)\right]V_{jn}$$

$$+\left[C_{22}^{ij}+D_{23}^{ji}+D_{23}^{ij}+F_{33}^{ij}+n^2C_{44}^{ij}-B_{55}^{ij}\frac{\partial^2}{\partial z^2}-\left(\omega^2+\Omega^2n^2\right)G^{ij}\right.$$

$$+\delta_{jN_{nr}}N_r\left(\frac{E_r A_r I_r n^4}{R_r^3}+\frac{E_r A_r}{R_r}+\frac{A_r d_r n^2}{R_r^2}-\frac{G_r J_r n^2}{R_r}\frac{\partial^2}{\partial z^2}\right)$$

$$+\delta_{jN_{nr}}\rho_r N_r A_r\Omega^2 R_r\left(\frac{I_r\left(n^4+n^2\right)}{R_r^2A_r}+1+\frac{2d_r n^2}{R_r}+n^2\right)$$

$$-\frac{\delta_{jN_{nr}}\rho_s N_s A_s}{\pi}\left(\frac{I_s n^2\omega^2}{A_s {R_s}^2}+\omega^2-\frac{2\Omega d_s n\omega}{R_s}+\frac{\Omega^2 I_s n^2}{A_s R_s^2}+\Omega^2\right)$$

$$\left.-\delta_{jN_{nr}}\rho_r N_r A_r\left(\frac{I_r n^2\omega^2}{A_r {R_r}^2}+\omega^2-\frac{2\Omega d_r n\omega}{R_r}+\frac{\Omega^2 I_r n^2}{A_r R_r^2}+\Omega^2\right)\right]W_{jn}$$

$$-\left[-A_{12}^{ij}-E_{13}^{ij}+E_{55}^{ji}+\frac{\delta_{jN_{nr}}\rho_s N_s A_s d_s\omega^2}{2\pi}+\delta_{jN_{nr}}\rho_r N_r A_r d_r\left(\frac{\omega^2-R_r n^3\Omega^2}{2}\right)\right]\frac{dU_{jn}}{dz}$$

$$+\left[\delta_{jN_{nr}}\left(-\frac{N_s G_s J_s n^2}{\pi R_s^2}+\frac{\rho_s N_s A_s I_s\omega^2}{\pi A_s}+\frac{\rho_r N_r A_r I_r}{A_r}\left(\omega^2-R_r n^2\Omega^2\right)\right)\right]\frac{d^2W_{jn}}{dz^2}$$

$$\left.-\left(\frac{\delta_{jN_{nr}}N_s E_s A_s d_s}{2\pi}\right)\frac{d^3U_{jn}}{dz^3}+\left(\frac{\delta_{jN_{nr}}N_s E_s I_s}{\pi}\right)\frac{d^4W_{jn}}{dz^4}\right\}=0.$$

The geometrical and natural boundary conditions at both ends of the shell are as follows:

either $$U_{in}\Big|_{z=0,L}=0$$

or $$\left[\left(A_{12}^{ij}+E_{13}^{ji}\right)W_{jn}+nA_{12}^{ij}V_{jn}+B_{11}^{ij}\frac{dU_{jn}}{dz}\right]\Bigg|_{z=0,L}=0\,;$$

either $$V_{in}\big|_{z=0,L}=0$$

or $$\left(-nA_{66}^{ij}U_{jn}+B_{66}^{ij}\frac{dV_{jn}}{dz}\right)\Bigg|_{z=0,L}=0\ ;$$

either $$W_{in}\big|_{z=0,L}=0$$

or $$\left(E_{55}^{ji}U_{jn}+B_{55}^{ij}\frac{dW_{jn}}{dz}\right)\Bigg|_{z=0,L}=0\ ,$$

where
$$A_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\psi_i\psi_j dr,\quad B_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\psi_i\psi_j rdr,\quad C_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\frac{\psi_i\psi_j}{r}dr,$$
$$D_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\psi_i'\psi_j dr,\quad E_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\psi_i'\psi_j rdr,$$
$$F_{kl}^{ij}=\int_{R_i}^{R_o}C_{kl}\psi_i'\psi_j' rdr,\quad G^{ij}=\int_{R_i}^{R_o}\rho\psi_i\psi_j rdr.$$

2.6. Differential quadrature discretization

The DQ method is numerical, with flexibility given for the arrangement of mesh points in the domain of interest. It also possesses higher-order interpolation characteristics. In order to use the DQ method to discretize the governing equations in the longitudinal direction z, each nodal surface is discretized into a set of N_z mesh points in this direction [30]. A brief review of the DQ method is presented in the Appendix A. Using DQ discretization rules at a mesh grid point in the domain and boundary conditions considered, the derivative of a sufficiently smooth function with respect to a coordinate direction can be approximated by taking a weighted linear sum of values of the function at all discrete mesh points in the coordinate direction. Thus, the partial derivatives of a function $f(z)$, as an example, at a point z_i are expressed as [32]

$$\frac{\partial^s f(z)}{\partial z^s}\Bigg|_{z=z_i}=\sum_{j=1}^{N_z}\tilde{C}_{ij}^s f(z_j),\quad i=1,2,\ldots,N_z,$$

where f can be taken to be U_{jn}, V_{jn}, or W_{jn}.

In the present paper, one of the following boundary conditions or some combinations of them is considered:

a) clamped
$$U_{in}=V_{in}=W_{in}=0\ ;$$
b) simply supported
$$V_{in}=W_{in}=N_{in}^z=0\ ;$$
c) free
$$N_{in}^z=M_{in}^{z\theta}=M_{in}^{zr}=0\ .$$

Thus, the whole system of differential equations has been discretized, and the global assembling leads to the set of linear algebraic equations
$$\{[M]\omega^2\}\{d\}+\{[G]\omega\}\{d\}+[K_{dd}]\{d\}+[K_{db}]\{b\}=0\ . \tag{14}$$

In the above equations, the vectors $\{d\}$ and $\{b\}$, with dimensions $3N_{nr}\left(N_z-2\right)$ and $6N_{nr}$, denote the unknowns at sampling points in the interior domain and at those on the boundary. The dimensions of $[M]$, $[K_{dd}]$, and $[G]$ are $N^*\times N^*\left[N^*=3N_{nr}\left(N_z-2\right)\right]$, but that of $[K_{db}]$ is $N^*\times 6N_{nr}$.

In a similar manner, the discretized form of the boundary conditions considered becomes

$$[K_{bd}]\{d\}+[K_{bb}]\{b\}=0, \tag{15}$$

where the dimension of $[K_{bd}]$ is $6N_{nr}\times N^*$ and that of $[K_{bb}]$ is $6N_{nr}\times 6N_{nr}$.

Using Eq. (15) to eliminate the boundary degrees of freedom $\{b\}$ from Eq. (14), it can be concluded that

$$\left\{[M]\omega^2\right\}\{d\}+\left\{[G]\omega\right\}\{d\}+\left\{[K_{dd}]-[K_{db}][K_{bb}]^{-1}[K_{bd}]\right\}\{d\}=0. \tag{16}$$

Relationship (16) is a nonstandard eigenvalue equation. For a given frequency, it can be transformed equivalently into the standard form [32]

$$\left(\underbrace{\begin{bmatrix}0 & I\\ -K & -G\end{bmatrix}}_{A^*}-\underbrace{\begin{bmatrix}I & 0\\ 0 & M\end{bmatrix}}_{B^*}\omega\right)\begin{Bmatrix}d\\ \omega d\end{Bmatrix}=0,$$

where I is the $N^*\times N^*$ identity matrix.

Applying the standard eigenproblem approach to the matrix $([B^*]^{-1}[A^*])$, $2N^*$ eigenvalues are obtained, from which two ones with the smallest absolute values are chosen. One of these eigenvalues is negative and corresponds to the backward wave, but the other one is positive and corresponds to the forward wave. In the case of a stationary cylindrical shell, these two eigenvalues are identical, and the vibration of the stiffened cylindrical shell is a standing wave motion.

3. Numerical Results and Discussion

In the presentation of results in figures, the backward and forward waves are shown as solid and dashed lines, respectively; the unit of the rotational speed Ω is rps (revolutions per second), and the natural frequencies is calculated for the in first longitudinal mode. In addition, five boundary conditions are considered here for the rotating stiffened cylindrical shell. These boundary conditions are the fully clamped (C-C), simply supported (S-S), simply supported at one edge and clamped at the other one (S-C), clamped at one edge and free at the other one (C-F), and simply supported at one edge and free at the other one (S-F). In addition, all layers are assumed to be of the same thickness. The material properties of the shell used in the present study are as follows: for the isotropic material — elastic modulus E = 7.6 GPa, Poisson ratio ν = 0.3, shear modulus 2.9 GPa, and density ρ = 1600 kg/m^3; for the orthotropic material — elastic moduli $E_{22}=E_{33}$ = 7.6 GPa and $E_{11}=2.5E_{22}$, Poisson ratios ν_{12} = 0.26 and $\nu_{13}=\nu_{23}$ = 0.2; shear moduli $G_{12}=G_{23}$ = 4.1 GPa and G_{13} = 2 GPa, and density ρ = 1643 kg/m^3. In addition, unless stated otherwise, for the stringers and ring stiffners, the following geometric and material parameters were used: thickness 8 mm, width 2 mm, elastic modulus E = 70 GPa, Poisson ratio ν = 0.3, and density ρ = 2707 kg/m^3.

Several examples are demonstrated to show the verification and efficiency of the results along with the convergence of the present analysis. The results obtained by the LW-DQ method are compared with those of other shell theories for stiffened and nonstiffened shells. A comparison of the results obtained in the new formulation with those found by Malekzadeh et al. [43] for laminated cylindrical shells shows an excellent agreement. Next, the effects of the number of stiffeners and rotational speed on the natural frequency of cylindrical shells are investigated.

TABLE 1. Convergence Behavior of the Frequency Parameter $f = \omega R\sqrt{\rho/E_{22}}$ of Rotating Stiffened Cylindrical Shells (L/R = 8, n = 8, [0/90], Ω =10 rev/s, F-C)

h/R	NMP	$N_1 = 2$		$N_1 = 4$		$N_1 = 6$	
		f_b	f_f	f_b	f_f	f_b	f_f
0.002	4	0.42394	0.41632	0.41271	0.40508	0.40983	0.40450
	5	0.39789	0.39026	0.39741	0.38978	0.39001	0.38947
	6	0.39147	0.38981	0.39008	0.38977	0.38951	0.38977
	7	0.39019	0.38986	0.38948	0.38983	0.38937	0.38983
	8	0.39058	0.38985	0.39049	0.38983	0.39045	0.38982
	9	0.39057	0.38985	0.39034	0.38982	0.39034	0.38982
	10	0. 39087	0.38984	0.39035	0.38982	0.39035	0.38982
	11	0. 39087	0.38984	0.39035	0.38982	0.39035	0.38982
0.02	4	0.53201	0.53201	0.51609	0.50793	0.52165	0.51350
	5	0.43509	0.42756	0.50257	0.49441	0.50222	0.49405
	6	0.40132	0.40132	0.24522	0.49443	0.49621	0.49409
	7	0.49856	0.49613	0.49671	0.49432	0.49646	0.49396
	8	0.49887	0.49606	0.49674	0.49433	0.49635	0.49399
	9	0.49862	0.49601	0.49681	0.49426	0.49630	0.493292
	10	0.49857	0.49597	0.49671	0.49427	0.49619	0.49393
	11	0.49856	0.49597	0.49670	0.49427	0.49619	0.49393

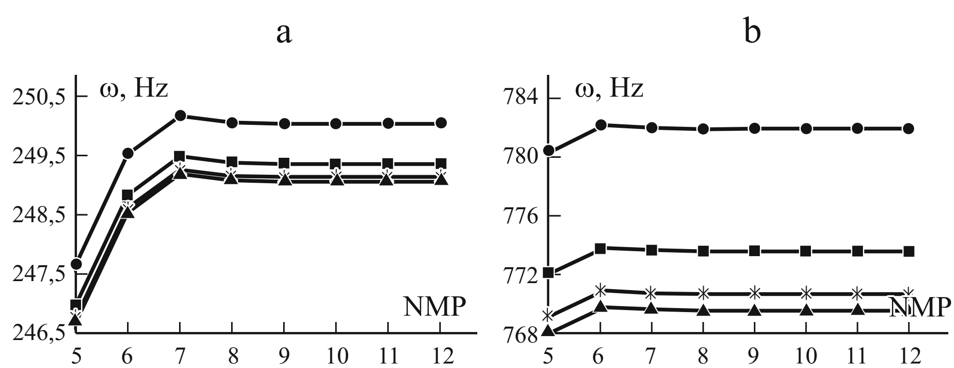

Fig. 2. Convergence behavior of the natural frequency ω of nonstiffened cylindrical shells (L/R = 4, n = 3, [0/90/0], S-S) with growing number of mesh points (NMP) at h/R = 0.1 (a) and 0.5 (b); N_l = 3 (●), 6 (■), 9 (∗), and 12 (▲).

3.1. Checking of convergence

In Table 3, the convergence behavior of the nondimensional fundamental frequency parameter $f = \omega R\sqrt{\rho / E_{22}}$ of forward and backward waves for a rotating stiffened cross-ply [0/90] laminated cylindrical shell with the F-C boundary condition is presented. The convergence behavior in relation to the number of mesh points (NMP) along the longitudinal direction and the number of mathematical layers through the thickness of the shell (N_l) were examined. Results were found for two different ratios of thickness to the mean radius of the shell. The fast rate of convergence of the method is quite evident.

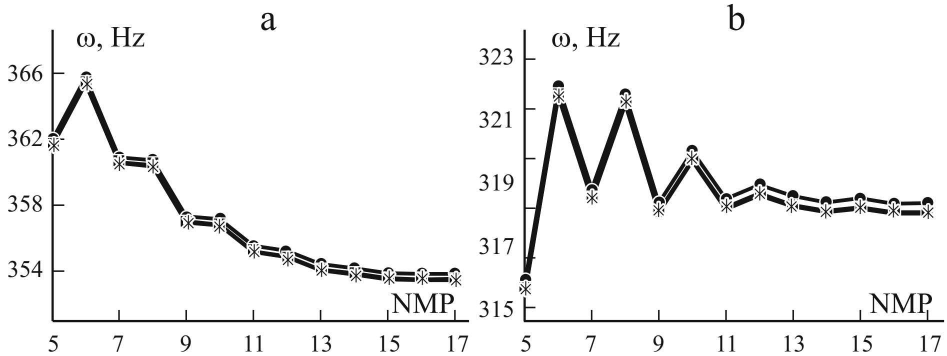

Fig. 3. Convergence behavior of the natural frequency ω of rotating stiffened cylindrical shells (L/R = 4, h/R = 0.05, n = 3, $N_s = N_r$ = 4, [0/90/0]) with C-C (a) and S-C (b) boundary conditions and N_l = 3 (●), 6 (□), and 9 (∗).

TABLE 2. Comparison of the frequency Parameters $f = \omega R\sqrt{\rho\left(1\text{-}\nu^2\right)/E}$ for Isotropic Laminated Cylindrical Shells (L/R = 1, $\nu = 0.3$, S-S)

h/R	h/L											
	0.1				0.2				0.4			
	n = 1		n = 2		n = 1		n = 2		n = 1		n = 2	
	[18]	Present	[18]	Present	[18]	Present	[18]	Present	[18]	Present	[18]	Present
0.2	0.5859	0.5849	0.4208	0.4223	0.9947	0.9921	0.9212	0.9253	2.0664	2.0415	2.1326	2.1432
0.4	0.2646	0.2629	0.3465	0.3417	0.6188	0.5916	0.5806	0.5769	1.2351	1.2216	1.2637	1.1832
0.6	0.1479	0.1475	0.4252	0.4051	0.4032	0.3965	0.5273	0.5114	0.9063	0.8929	0.9501	0.9328

The convergence behavior of the method for nonrotating nonstiffened cylindrical shells with two different thicknesses is presented in Fig. 2. As it is well seen from these figures, with increasing thickness, the convergence is postponed.

Lastly, the convergence behavior of the method for rotating stiffened cylindrical shells with two different boundary conditions, C-C and S-C, are presented in Fig. 3. It is clearly visible that the convergence behavior of the stiffened shell is postponed, whereas the results for the nonstiffened shell converge rapidly. As seen, converging results are achieved with NMP = 8 and 16 for the nonstiffened and stiffened shells, respectively. In addition, the results presented show that the convergence for the stiffened cylindrical shell with different boundary conditions behaves differently.

3.2. Validation

In order to validate the results obtained, first, the formulation was performed for an isotropic cylindrical shell with simply supported boundary conditions. The results for the first two frequency parameters $f = \omega R\sqrt{\rho\left(1-\nu^2\right)/E}$ of thick cylindrical shells, listed in Table 2, are compared with those of LW-exact solutions found by Loy and Lam [18] at different values of h/L and h/R.

Further comparisons are demonstrated with the work done by Khdeir et al. [7], which considered different theories (HSDT, FSDT, and CST) for nonrotating cylindrical shells. The values of the fundamental frequency parameter $f = \left(\omega L^2/100h\right)\sqrt{\rho/E_{22}}$ of symmetric and antisymmetric cross-ply laminated cylindrical shells under various boundary conditions are listed in Table 3. The trend of the results seen in this table points to the accuracy of the present work, as it is based on the 3-D theory of elasticity.

TABLE 3. Comparison of the Fundamental Frequency Parameters $f = \left(\omega L^2 / 100h\right)\sqrt{\rho / E_{22}}$ for Cross-Ply Laminated Cylindrical Shells under Various Boundary Conditions (h/R = 0.2, L/R = 2, m = 1)

Theory	S-S	S-C	C-C	C-F
		[0/90]		
CST [7]	0.1630	0.1841	0.2120	0.0938
FSDT [7]	0.1552	0.1697	0.1876	0.0914
HSDT [7]	0.1566	0.1726	0.1928	0.0921
LW-DQ (present)	0.1371	0.1531	0.1718	0.0909
		[0/90/0]		
CST [7]	0.2073	0.2662	0.3338	0.1099
FSDT [7]	0.1779	0.1945	0.2129	0.0988
HSDT [7]	0.1777	0.1972	0.2191	0.0995
LW-DQ (present)	0.1729	0.1903	0.2108	0.0992
		$[0/90]_5$		
CST [7]	0.1958	0.2304	0.2752	0.1077
FSDT [7]	0.1899	0.2012	0.2137	0.1004
HSDT [7]	0.1900	0.2027	0.2166	0.1010
LW-DQ (present)	0.1802	0.1977	0.2104	0.1379

TABLE 4. Comparison of Natural Frequencies of Stiffened Cross-Ply Laminated Cylindrical Shells (L/R = 4, h/R = 0.005, E_{11}/E_{22} = 2.5, ν_{12} = 0.26, [0/90/0])

n	Stringer/ring configuration					
	2/2			4/4		
	[40]	[42]	Present	[40]	[42]	Present
1	568.0	568.9	573.0	561.4	562.0	570.1
2	278.1	278.8	290.0	286.4	287.2	307.2
3	216.2	216.9	223.2	261.4	261.8	267.6
4	306.7	307.9	306.2	404.5	405.8	391.5
5	469.4	470.7	464.0	627.5	629.4	599.2
6	676.9	679.5	667.7	906.9	909.9	863.2
7	924.1	927.2	910.9	1238.4	1241.5	1177.6
8	1209.7	1212.8	1192.0	1620.9	1624.4	1540.6
9	1533.3	1537.1	1510.7	2053.9	2059.0	1951.8
10	1894.7	1899.8	1866.5	2537.4	2544.0	2410.7

The third comparison, shown in Table 4, is for a nonrotating stiffened laminated cylindrical shell with simply supported boundary conditions at $\Omega = 0$ in the present formulations and in those of the exact method obtained by Zhao et al. [40] and Talebitooti et al.[42].

The last comparison, presented in Fig. 4, is for a rotating stiffened laminated cylindrical shell with simply supported boundary condition in the present formulation and in that of exact method considered by Talebitooti et al. [42]. As, can be seen, the two sets of frequencies are very close to each other, but the present results are a bit lower than those found in [42]. This may be due to the difference between the shell theories. The classic shell theory has been employed in [42]. In this paper, the 3-D shell theory is used.

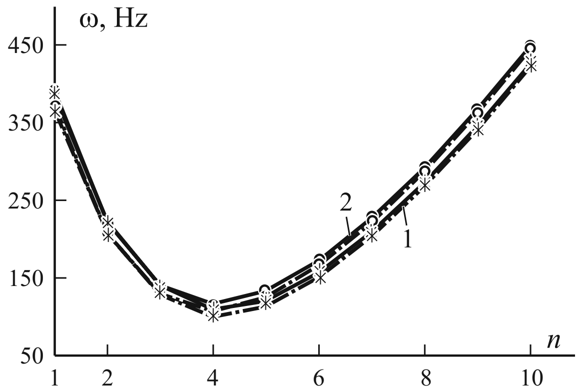

Fig. 4. Variation in the natural frequency ω with growing number n of circumferential waves for a rotating stiffened composite cylindrical shell ($L/R = 2$, $h/R = 0.002$, $N_s = N_r = 10$, $\Omega = 10$ (rps), [0/90/0], S-S)according to the present study (1) and [42] (2).

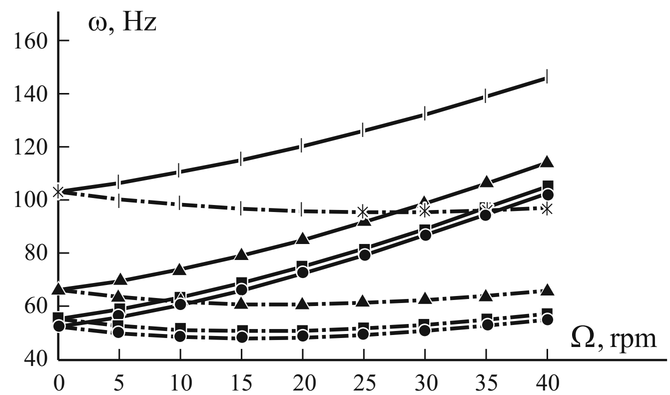

Fig. 5. Variation in the natural frequencies ω of a stiffened composite cylindrical shell ($L/R = 6$, $h/R = 0.002$, $n = 3$, $N_s = N_r = 10$, [45/–45/45], F-C) with growing rotational speed Ω at different depth to width ratios $d_{s(\mathrm{or}\,r)}/b_{s(\mathrm{or}\,r)}$: 1 (●), 2 (■), 4 (▲), and 8 (✳).

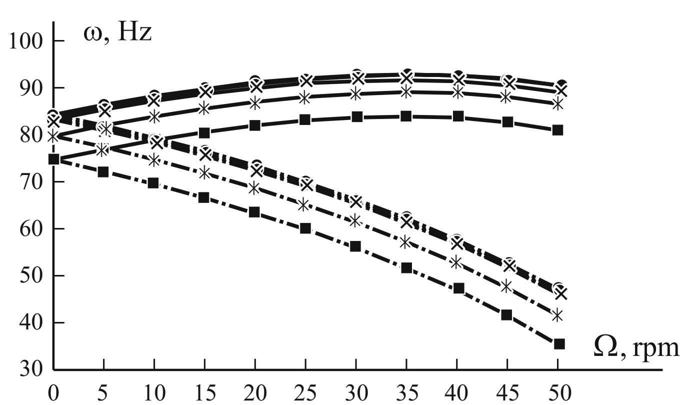

Fig. 6. Variation in the natural frequencies ω of a stiffened cylindrical shell ($L/R = 4$, $h/R = 0.002$, $n = 4$, $N_r = 0$, [45/–45/45], F-C) with growing rotational speed Ω at the numbers of stringers $N_s = 10$ (■), 20 (✳), 40 (·), 60 (○), and 80 (×).

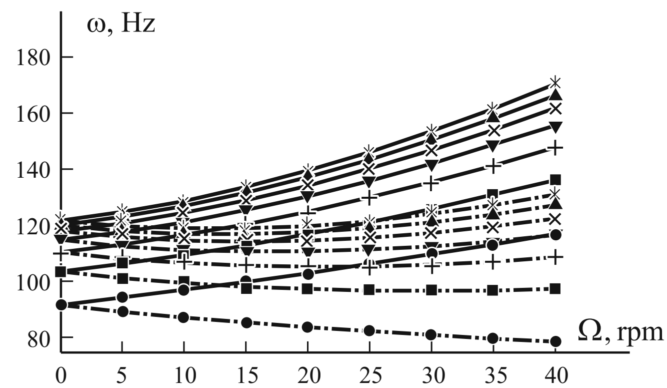

Fig. 7. Variation in the natural frequencies ω for a stiffened cylindrical shell ($L/R = 4$, $h/R = 0.002$, $n = 4$, $N_s = 0$, [45/–45/45], F-C) with growing rotational speed Ω at the numbers of rings $N_r = 5$ (●), 10 (■), 15 (x), 20 (▼), 25 (×), 30 (▲), and 35 (✳).

3.3. Parameter studies

The previous sections demonstrate the fast convergence behavior and adaptability of the present method. Here, the effects of depth to width ratio and the number of stiffeners on the natural frequency of the stiffened cylindrical shells are considered.

Figure 5 illustrates the effects of the depth to width ratio of stiffeners on the natural frequencies of rotating stiffened cylindrical shell at different rotational speeds if the cross-sectional area of the stiffeners is constant. It is observed that, at different rotating speeds, the frequencies of the shells generally increase with increasing depth to width ratio of the stiffeners both for forward and backward waves. It is also seen that, at high ratios, the effects of increasing depth to width ratio on frequencies of the shell are more pronounced.

Figure 6 depicts the effects of variation in the number of stringers on the frequencies characteristic of shells. As may be seen, the natural frequency grows as the number of stringers increases. However, at high rotational speeds, the frequency slightly decreases with increasing number of stringers. This is due to the fact that, in the first longitudinal mode (i.e., at low frequencies), the inertial terms for stiffened shells are more significant than their stiffness.

Figure 7 shows effects of the number of rings on the frequencies of rotating cylindrical shells. As viewed, the frequencies of the shells generally increase with increasing number of rings both for forward and backward waves, whereas the increase in the gradient rate becomes small.

4. Conclusion

In this paper, a three-dimensional analysis of free vibration of thick rotating stiffened composite cylindrical shells is developed using a mixed layerwise-differential quadrature method. Hamilton's principle with the layerwise theory is used to discretize the through-thickness form of the equation of motion and the related boundary conditions of stiffened cylindrical shells. Then, the equations of motion and of boundary conditions are transformed into a set of algebraic equations by using the DQM in the longitudinal direction. It is found that this approach leads to a fast rate of convergence and yields accurate results compared with those available in the literature. It is concluded that the mixed layerwise-differential quadrature method can be used as a robust, effective, and accurate numerical way for analyzing thick rotating stiffened cylindrical shells. In addition, the stiffeners in a cylindrical shell increases its natural frequency. However, the effect of stringers on the natural frequency in the first longitudinal mode, depending on the rotational speed of the shell, is not pronounced. At high depth to width ratios of the stiffeners, the effects of increase in this ratio on the natural frequencies of the shell are more pronounced

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Appendix A. The basic idea of the DQ method

The DQM is based on the simple mathematical concept that any sufficiently smooth function in a domain can be expressed approximately as an $(N-1)$th-order polynomial in the domain. In other words, at a discrete mesh point in a domain, the partial derivatives of a function $f(z)$ at a point (z_i) are expressed as [32]

$$\left.\frac{\partial^s f(z)}{\partial z^s}\right|_{z=z_i} = \sum_{j=1}^{N} \tilde{C}_{ij}^s f\left(z_j\right), \quad i = 1, 2, ..., N,$$

where N is the number of mesh points, and f can be taken to be u, v and w in the present study. $\tilde{C}_{ij}^{s}$ are the respective weighting coefficients related to an sth-order derivative and is obtained as follows.

If $s = 1$, namely, for the first-order derivative,

$$\tilde{C}_{ij}^{1} = \frac{M^{(1)}\left(z_i\right)}{\left(z_i - z_j\right)M^{(1)}\left(z_j\right)} \text{ for } i \neq j \text{ and } i, j = 1, 2, \ldots, N,$$

and

$$\tilde{C}_{ii}^{1} = -\sum_{j=1(j\neq i)}^{N} \tilde{C}_{ij}^{1} \text{ for } i, j = 1, 2, \ldots, N,$$

where $M^{(1)}(z)$ is the first derivative of $M(z)$, and they can be defined as

$$M\left(z\right) = \prod_{j=1}^{N}\left(z - z_j\right), \quad M^{(1)}\left(z_k\right) = \prod_{j=1(j\neq k)}^{N}\left(z_k - z_j\right).$$

If $s > 1$, namely, for the second and higher-order derivatives, the weighting coefficient are obtained using the following simple recurrence relationships:

$$\tilde{C}_{ij}^{s} = r\left(\tilde{C}_{ij}^{1} \cdot \tilde{C}_{ii}^{s-1} - \frac{\tilde{C}_{ij}^{s-1}}{z_i - z_j}\right) \text{ for } i \neq j \text{ and } i, j = 1, 2, \ldots, N, \; s = 2, 3, \ldots, N-1$$

$$\tilde{C}_{ii}^{s} = -\sum_{j=1(j\neq i)}^{N} \tilde{C}_{ij}^{s}, \; i = 1, 2, \ldots, N.$$

Since the coordinate distribution and the number of discrete mesh points (NMP) can be chosen arbitrarily in implementation of the DQM, the following distributions of the mesh points in the longitudinal direction z are used in the present formulation:

$$z_i = \frac{i-1}{N-1}L, \; i = 1, 2, \ldots, N.$$